ECEGR 5760, Winter 2026
A Series of Lab Explorations on Bayesian Inference and Markov-chain Monte Carlo Simulation,
Part 2

What You Will Do

In this lab, we will focus on the Bayesian hyperparameter (really more of a meta-parameter) kappa followed by Bayesian modeling for regression.

Why

These and subsequent Bayesian methods of inference are the explicit bases of robotics AI (from the Kálmán and particle filters to SLAM) as well as implicitly forming the basis for transformers.

Procedures and Background Information

This series of lab handouts includes the necessary theory for the lab exercises. Some of the information may be review for you. It is included in order to make the documents self-sufficient (with complete explanations of both theory and methods).

Step 1

The Bayesian viewpoint turned the parameter r into a random variable. In working with this random variable, we saw that—at least for some problems—the family of beta functions, parametrized by α and β, generate intuitively appropriate curves.

We also saw that these parameters can encode real and imagined previous binomial trials: We can set our fair-coin-scenario parameters as $\alpha = \beta = 1$ or $\alpha = \beta = 50$ or $\alpha = \beta = 1000$ or $\alpha = \beta = 2$, and so on. Whenever $\alpha = \beta$, we are setting up the fair-coin assumption as the prior, but the larger the value, the more confident we are in our assumption of coin fairness.

Notice how, in the previous lab, the first and third scenarios had small values for α and β. This is why they converged on the evidence provided by new tosses sooner than the fair-coin scenario.

We could also have indicated greater confidence in the coin being biased by setting the two parameters to larger values at the same proportion or even choose a different proportion to indicate our belief in greater and smaller degrees of bias in the coin.

Thus, we can see α and β also as random variables.

In this case, we need to define a prior density over all our random variables:

$$p(r, \alpha, \beta) = p(r|\alpha, \beta)p(\alpha, \beta)$$

For simplicity, let's assume α and β are independent. (As our belief in a given scenario becomes stronger, this is less true, so let's assume meta-ignorance for the moment: We are not sure whether we have no prior belief about the coin or whether we do expect it to be fair or to be biased.)

Moving towards a Solution

The quantity we want to find is the posterior distribution over all parameters

$$p(r, \alpha, \beta|y_N) = \frac{p(y_N|r, \alpha, \beta)p(r, \alpha, \beta)}{p(y_N)}$$

Next, we use conditional independence to simplify this:

$$p(r, \alpha, \beta|y_N) = \frac{p(y_N|r)p(r, \alpha, \beta)}{p(y_N)}$$

The distribution depends on α and β, but only through their influence on r: Given a value for r, y_N is not further dependent on α or β.

And, finally, substituting $p(r, \alpha, \beta) = p(r|\alpha, \beta)p(\alpha, \beta)$ into the above:

$$p(r, \alpha, \beta|y_N) = \frac{p(y_N|r)p(r|\alpha, \beta)p(\alpha, \beta)}{p(y_N)}$$

$p(\alpha, \beta)$ requires a parameter that controls this prior density in the same way that α and β control the density for r:

$$p(\alpha, \beta) = p(\alpha, \beta|\kappa)$$

κ (kappa) is the hyperparameter controlling the prior on the parameters controlling the prior on r.

$$p(r, \alpha, \beta|y_N, \kappa) = \frac{p(y_N|r)p(r|\alpha, \beta)p(\alpha, \beta|\kappa)}{p(y_N|\kappa)}$$

where

$$p(y_N|\kappa) = \iiint p(y_N|r)p(r|\alpha,\beta)p(\alpha,\beta|\kappa)\, d\beta\, d\alpha\, dr$$

Observations:

- This usually becomes intractable (cannot be computed) at this level.
- Whether we go to this extent (or even further!) depends on our understanding of the dataset and the problem domain. We might choose to hyperparametrize further before fixing any parameters or we might choose to fix κ (or even α and β).
- Such logical spirals are not limited to Bayesian inference; frequentist statistics has the same burden, just in other issues (definitions, questions, ...), for example, in how the probability of H for a fair coin does not guarantee one heads and one tails for every two tosses or two heads or two tails for every four tosses and even for 200,000 tosses, 100,000 heads simply becomes very likely but not guaranteed. Similarly, a confidence interval around a quantity is itself probabilistic. There can be a confidence interval around the probability of such a confidence interval, and so forth. Fuzziness continues down the hierarchy to infinity. Mathematical probability is a human invention.

Bayesian Regression

Table 1 shows the gold-, silver-, and bronze-medal-winning times from men's 100-meter races in the Olympics, 1896 through 2008.

1896	12	12.2	12.6
1900	11	11.1	11.2
1904	11	11.2	11.2
1906	11.2	11.3	11.3
1908	10.8	10.9	11
1912	10.8	10.9	10.9
1920	10.8	10.8	11
1924	10.6	10.7	10.8
1928	10.8	10.9	10.9
1932	10.3	10.3	10.4
1936	10.3	10.4	10.5
1948	10.3	10.4	10.4
1952	10.4	10.4	10.4
1956	10.5	10.5	10.6
1960	10.2	10.2	10.3
1964	10	10.2	10.2
1968	9.95	10	10
1972	10.14	10.24	10.33
1976	10.06	10.08	10.14
1980	10.25	10.25	10.39

1984	9.99	10.19	10.22
1988	9.92	9.97	9.99
1992	9.96	10.02	10.04
1996	9.84	9.89	9.9
2000	9.87	9.99	10.04
2004	9.85	9.86	9.87
2008	9.69	9.89	9.91

To predict future times, we need to fit a curve to the data. In the Bayesian approach to regression, as before, we define the prior and the likelihood and use them to compute the posterior density over the parameters of our model.

Plotting the data will suggest that the relationship is not a line, so we will use a polynomial and additive zero-mean Gaussian noise:

$$t_n = w_0 + w_1 x_n + w_2 {x_n}^2 + \cdots + w_K {x_n}^K + \epsilon_n$$
$$= \mathbf{w}^{\mathrm{T}} \mathbf{x}_n + \epsilon_n$$

where

$$\mathbf{x}_n = [1, x_n, {x_n}^2, \dots, {x_n}^K]^{\mathrm{T}}$$
$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]^{\mathrm{T}}$$
$$\mathbf{w} = [w_0, \dots, w_K]^{\mathrm{T}}$$
$$\mathbf{t} = [t_1, \dots, t_N]^{\mathrm{T}}$$
$$\boldsymbol{\epsilon} = [\epsilon_1, \dots, \epsilon_N]^{\mathrm{T}}$$

such that

$$\mathbf{t} = \mathbf{X}\mathbf{w} + \boldsymbol{\epsilon}$$

Normally, we would not know the variance of the additive noise. We will simplify the work by assuming a value for the variance. The resulting calculations can then serve as a model for when we do not make assumptions about variance either.

$$p(\mathbf{w}|\mathbf{t}, \mathbf{X}, \sigma^2, \Delta) = \frac{p(\mathbf{t}|\mathbf{w}, \mathbf{X}, \sigma^2, \Delta) p(\mathbf{w}|\Delta)}{p(\mathbf{t}|\mathbf{X}, \sigma^2, \Delta)}$$

(where Δ is some set of parameters necessary for defining the prior over $\mathbf{w}$)

$$p(\mathbf{w}|\mathbf{t},\mathbf{X},\sigma^2,\Delta) = \frac{p(\mathbf{t}|\mathbf{w},\mathbf{X},\sigma^2,\Delta)p(\mathbf{w}|\Delta)}{\oint p(\mathbf{t}|\mathbf{w},\mathbf{X},\sigma^2,\Delta)p(\mathbf{w}|\Delta)\ d\mathbf{w}}$$

Working out the Likelihood

$\boldsymbol{\epsilon}$ is a zero-mean Gaussian. $\mathbf{Xw}$ is a constant. Their sum is an N-dimensional Gaussian density with the same variance but with mean equal to $\mathbf{Xw}$:

$$p(\mathbf{t}|\mathbf{w},\mathbf{X},\sigma^2,\Delta) = p(\mathbf{t}|\mathbf{w},\mathbf{X},\sigma^2) = \mathcal{N}(\mathbf{Xw},\sigma^2\mathbf{I}_N)$$

Working out the Prior

In order to be able to do the calculations, we choose a prior that is conjugate to the likelihood. The conjugate for a Gaussian likelihood is a Gaussian prior, so the prior for $\mathbf{w}$ is a Gaussian with unknown mean and variance:

$$p(\mathbf{w}|\Delta \equiv \{\boldsymbol{\mu}_0,\boldsymbol{\Sigma}_0\}) = \mathcal{N}(\boldsymbol{\mu}_0,\boldsymbol{\Sigma}_0)$$

Computing the Posterior

The trick is to ignore the marginal likelihood (the evidence, the normalizing factor) and to use algebra on the product of the likelihood and the prior until we get a Gaussian form (with some constant multiplier).

As part of this, we collect terms involving $\mathbf{w}$ and ignore terms that are not a function of $\mathbf{w}$.

$$p(\mathbf{w}|\mathbf{t},\mathbf{X},\sigma^2,\Delta) \propto p(\mathbf{t}|\mathbf{w},\mathbf{X},\sigma^2)p(\mathbf{w}|\boldsymbol{\mu}_0,\boldsymbol{\Sigma}_0) = \mathcal{N}(\mathbf{Xw},\sigma^2\mathbf{I}_N) \times \mathcal{N}(\boldsymbol{\mu}_0,\boldsymbol{\Sigma}_0)$$

$$= \frac{1}{2\pi^{N/2}\ |\sigma^2\mathbf{I}_N|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{t}-\mathbf{Xw})^{\mathrm{T}}(\sigma^2\mathbf{I}_N)^{-1}\ (\mathbf{t}-\mathbf{Xw})\right)$$

$$\times \frac{1}{2\pi^{N/2}\ |\boldsymbol{\Sigma}_0|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{w}-\boldsymbol{\mu}_0)^{\mathrm{T}}(\boldsymbol{\Sigma}_0)^{-1}\ (\mathbf{w}-\boldsymbol{\mu}_0)\right)$$

(That's just substituting our Gaussians' parameters into the gaussian equation.)

Next, we remove terms that aren't functions of $\mathbf{w}$.

$$p(\mathbf{w}|\mathbf{t},\mathbf{X},\sigma^2,\Delta) \propto \exp\left(-\frac{1}{2\sigma^2}(\mathbf{t}-\mathbf{Xw})^{\mathrm{T}}\ (\mathbf{t}-\mathbf{Xw})\right) \times \exp\left(-\frac{1}{2}(\mathbf{w}-\boldsymbol{\mu}_0)^{\mathrm{T}}(\boldsymbol{\Sigma}_0)^{-1}\ (\mathbf{w}-\boldsymbol{\mu}_0)\right)$$

$$= \exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma^2}(\mathbf{t}-\mathbf{Xw})^{\mathrm{T}}\ (\mathbf{t}-\mathbf{Xw}) + (\mathbf{w}-\boldsymbol{\mu}_0)^{\mathrm{T}}(\boldsymbol{\Sigma}_0)^{-1}\ (\mathbf{w}-\boldsymbol{\mu}_0)\right)\right\}$$

$$\propto \exp\left\{-\frac{1}{2}\left(-\frac{2}{\sigma^2}\mathbf{t}^{\mathrm{T}}\mathbf{Xw} + \frac{1}{\sigma^2}\mathbf{w}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{Xw} + \mathbf{w}^{\mathrm{T}}{\boldsymbol{\Sigma}_0}^{-1}\mathbf{w} - \mathbf{2}{\boldsymbol{\mu}_0}^{\mathrm{T}}{\boldsymbol{\Sigma}_0}^{-1}\mathbf{w}\right)\right\}$$

where the last line results from multiplying out the various matrix products and leaving out any terms that do not have $\mathbf{w}$ in them.

We know, from conjugate pairs, that the above will equal another Gaussian density: $\mathcal{N}(\boldsymbol{\mu}_\mathbf{w}, \boldsymbol{\Sigma}_\mathbf{w})$, which when written out explicitly appears as follows.

$$p(\mathbf{w}|\mathbf{t}, \mathbf{X}, \sigma^2, \Delta) = \mathcal{N}(\boldsymbol{\mu}_\mathbf{w}, \boldsymbol{\Sigma}_\mathbf{w}) \propto \exp\left(-\frac{1}{2}(\mathbf{w} - \boldsymbol{\mu}_\mathbf{w})^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}(\mathbf{w} - \boldsymbol{\mu}_\mathbf{w})\right)$$

which is in turn proportional to the further simplified product (missing the mean-squared term):

$$\exp\left(-\frac{1}{2}\mathbf{w}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w} - 2\boldsymbol{\mu}_\mathbf{w}{}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w}\right)$$

The arguments of the two exponentials (the proposed posterior and the calculated posterior) have to be equal:

$$-\frac{1}{2}\mathbf{w}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w} - 2\boldsymbol{\mu}_\mathbf{w}{}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w} = -\frac{1}{2}\left(-\frac{2}{\sigma^2}\mathbf{t}^\mathrm{T}\mathbf{X}\mathbf{w} + \frac{1}{\sigma^2}\mathbf{w}^\mathrm{T}\mathbf{X}^\mathrm{T}\mathbf{X}\mathbf{w} + \mathbf{w}^\mathrm{T}\boldsymbol{\Sigma}_0{}^{-1}\mathbf{w} - 2\boldsymbol{\mu}_0{}^\mathrm{T}\boldsymbol{\Sigma}_0{}^{-1}\mathbf{w}\right)$$

Simplifying further:

$$\mathbf{w}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w} - 2\boldsymbol{\mu}_\mathbf{w}{}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w} = \left(-\frac{2}{\sigma^2}\mathbf{t}^\mathrm{T}\mathbf{X}\mathbf{w} + \frac{1}{\sigma^2}\mathbf{w}^\mathrm{T}\mathbf{X}^\mathrm{T}\mathbf{X}\mathbf{w} + \mathbf{w}^\mathrm{T}\boldsymbol{\Sigma}_0{}^{-1}\mathbf{w} - 2\boldsymbol{\mu}_0{}^\mathrm{T}\boldsymbol{\Sigma}_0{}^{-1}\mathbf{w}\right)$$

Let's match up the quadratic ($\mathbf{w}$-squared) terms:

$$\mathbf{w}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w} = \frac{1}{\sigma^2}\mathbf{w}^\mathrm{T}\mathbf{X}^\mathrm{T}\mathbf{X}\mathbf{w} + \mathbf{w}^\mathrm{T}\boldsymbol{\Sigma}_0{}^{-1}\mathbf{w} \quad (\mathbf{1})$$

And let's match up the linear terms:

$$-2\boldsymbol{\mu}_\mathbf{w}{}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w} = -\frac{2}{\sigma^2}\mathbf{t}^\mathrm{T}\mathbf{X}\mathbf{w} - 2\boldsymbol{\mu}_0{}^\mathrm{T}\boldsymbol{\Sigma}_0{}^{-1}\mathbf{w}$$

or:

$$\boldsymbol{\mu}_\mathbf{w}{}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w} = \frac{1}{\sigma^2}\mathbf{t}^\mathrm{T}\mathbf{X}\mathbf{w} + \boldsymbol{\mu}_0{}^\mathrm{T}\boldsymbol{\Sigma}_0{}^{-1}\mathbf{w} \quad (\mathbf{2})$$

Collecting like terms in equation $(\mathbf{1})$:

$$\mathbf{w}^\mathrm{T}\boldsymbol{\Sigma}_\mathbf{w}{}^{-1}\mathbf{w} = \mathbf{w}^\mathrm{T}\left(\frac{1}{\sigma^2}\mathbf{X}^\mathrm{T}\mathbf{X} + \boldsymbol{\Sigma}_0{}^{-1}\right)\mathbf{w}$$

meaning:

$$\boldsymbol{\Sigma}_\mathbf{w}{}^{-1} = \frac{1}{\sigma^2}\mathbf{X}^\mathrm{T}\mathbf{X} + \boldsymbol{\Sigma}_0{}^{-1}$$

or, equivalently,

$$\boldsymbol{\Sigma}_{\mathbf{w}} = \left(\frac{1}{\sigma^2}\mathbf{X}^{\mathrm{T}}\mathbf{X} + \boldsymbol{\Sigma}_0{}^{-1}\right)^{-1} \qquad (\mathbf{3})$$

And for the linear terms (from equation $(\mathbf{2})$), we start by eliminating $\mathbf{w}$ by multiplying through by $\mathbf{w}^{-1}$:

$$\boldsymbol{\mu}_{\mathbf{w}}{}^{\mathrm{T}}\boldsymbol{\Sigma}_{\mathbf{w}}{}^{-1} = \frac{1}{\sigma^2}\mathbf{t}^{\mathrm{T}}\mathbf{X} + \boldsymbol{\mu}_0{}^{\mathrm{T}}\boldsymbol{\Sigma}_0{}^{-1}$$

And then, to find the mean, we do same with the covariance matrix of the Gaussian:

$$\boldsymbol{\mu}_{\mathbf{w}}{}^{\mathrm{T}}\boldsymbol{\Sigma}_{\mathbf{w}}{}^{-1}\boldsymbol{\Sigma}_{\mathbf{w}} = \left(\frac{1}{\sigma^2}\mathbf{t}^{\mathrm{T}}\mathbf{X} + \boldsymbol{\mu}_0{}^{\mathrm{T}}\boldsymbol{\Sigma}_0{}^{-1}\right)\boldsymbol{\Sigma}_{\mathbf{w}} = \boldsymbol{\mu}_{\mathbf{w}}{}^{\mathrm{T}}$$

$$\boldsymbol{\mu}_{\mathbf{w}} = \boldsymbol{\Sigma}_{\mathbf{w}}\left(\frac{1}{\sigma^2}\mathbf{t}^{\mathrm{T}}\mathbf{X} + \boldsymbol{\mu}_0{}^{\mathrm{T}}\boldsymbol{\Sigma}_0{}^{-1}\right)$$

where the last step is because covariance matrices are symmetric (and where $\boldsymbol{\Sigma}_{\mathbf{w}}$ refers to $(\mathbf{3})$).

We had previously said that $p(\mathbf{w}|\mathbf{t}, \mathbf{X}, \sigma^2, \Delta) = \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}}, \boldsymbol{\Sigma}_{\mathbf{w}})$. We now know how to write $\mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}}, \boldsymbol{\Sigma}_{\mathbf{w}})$. in terms of $\mathbf{t}$ (the target values from the training set), $\mathbf{X}$ (the stack of feature vectors from the training set), $\boldsymbol{\mu}_0$ and $\boldsymbol{\Sigma}_0$ (the mean and covariance of our prior density), and σ^2 (the noise variance, for which we will pick a conservative—large—value).

The resulting equation is repeated here for clarity:

$$p(\mathbf{w}|\mathbf{t}, \mathbf{X}, \sigma^2, \Delta) = \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}}, \boldsymbol{\Sigma}_{\mathbf{w}})$$

$$\boldsymbol{\Sigma}_{\mathbf{w}} = \left(\frac{1}{\sigma^2}\mathbf{X}^{\mathrm{T}}\mathbf{X} + \boldsymbol{\Sigma}_0{}^{-1}\right)^{-1}$$

$$\boldsymbol{\mu}_{\mathbf{w}} = \boldsymbol{\Sigma}_{\mathbf{w}}\left(\frac{1}{\sigma^2}\mathbf{t}^{\mathrm{T}}\mathbf{X} + \boldsymbol{\mu}_0{}^{\mathrm{T}}\boldsymbol{\Sigma}_0{}^{-1}\right)$$

The regularized least-squares solution for the same regression problem in the standard approach gives

$$\widehat{\mathbf{w}} = \left(\mathbf{X}^{\mathrm{T}}\mathbf{X} + N\lambda\mathbf{I}\right)^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{t}$$

which is of the same form as our posterior mean. That mean and the standard solution are what is called the maximum-*a posteriori* estimate (MAP estimate).

The MAP estimate is the maximum-likelihood estimate informed by the prior and is thus a Bayesian*ish* estimate. The full Bayesian estimate is the complete posterior distribution.

Doing Prediction with Our Results

Recall that making a single prediction means taking the expectation of the posterior density, where the winning time t_{new} results from some x_{new}.

Let's use a first-order polynomial (a line) so that densities can be visualized in two dimensions.

$$\mathbf{x}' = \frac{\mathbf{x} - x_1}{4}$$

(Subtracting 1896 from each Olympics year and dividing by 4 turns the Olympics years into simple index values.)

$$\mathbf{x}'_n = [1, x'_n]^{\mathrm{T}}$$

Deliverable (Work):

Print the rescaled men's-100-meter-race data.

Drawing Regression Lines from the Posterior:

A good way to express ignorance about the mean-parameter vector $\boldsymbol{\mu}_0$ is to set both prior means to zero: $\boldsymbol{\mu}_0 = [0, 0]^{\mathrm{T}}$.

For the covariance, since zero is not an option, say we recall w_0 being much higher than w_1 in the standard regression solution for the 100-m data. (Linear regression on the data gives the best-fit line of $f(x;\ w_0, w_1) = 36.416 - 0.013x$. In other words, w_0 needs to be able to swing a lot further than w_1. Using this information may qualify for the Texas-sharpshooter fallacy; we do so in order to observe how a distribution function on regression parameters can approach the linear-regression result. With less well-informed priors, we simply would need a lot more data, but—given either informed priors or a lot of data—Bayesian inference performs very well. And a lot of data is no longer too much to ask.)

Given the three-orders-of-magnitude difference, selecting some round numbers as in the following, with only two orders of magnitude difference, should only be moderately questionable.

$$\boldsymbol{\Sigma}_0 = \begin{bmatrix} 100 & 0 \\ 0 & 5 \end{bmatrix}$$

Note that the diagonal elements are zero because we made the naive assumption of independence in the prior. (This does not preclude them from turning out dependent in the posterior.)

To make up for our moderately helpful prior covariance, we will choose a largish noise variance to make convergence a little less easy for the posterior: $\sigma^2 = 10$

Deliverable (Work):

Either use directly or port to another language the MATLAB code *olympbayes.m* and interpret the plots that result: What do the density functions given by concentric equipotentials represent and how are they changing? What do the various line segments represent? (What do they result from? Why do they vary as they do? How does their degree of variation change) What does the final density plot tell you?

Experiment with the variable 'ss' and deduce what it is and what effect it has on the process.